



USE CASE MODELING

NeuroForce SDLC Platform

Introduction

UML & Use Case Modeling — the basics



What is UML?

Unified Modeling Language — a standard visual language for specifying, constructing, and documenting software systems.



What is a Use Case Diagram?

A UML behavioral diagram showing actors, system functions (use cases), and their relationships within a system boundary.



Why Use Case Modeling?

Captures functional requirements from a user's perspective before coding begins, aligning stakeholders and developers.



Benefits in SDLC

Clarifies scope, improves communication, supports test-case design, and reduces rework across the SDLC.

About the Project

NeuroForce SDLC Platform



Objective & Purpose

NeuroForce is a web-based Software Development Life Cycle (SDLC) management platform that helps teams plan, track, and deliver software projects. It unifies project planning, backlog management, development tracking, and reporting into one system.



Who Uses It?

- Admin — configures the platform
- Project Manager — plans & tracks delivery
- Project Developer — builds & reports progress

Main Features



User, role & permission administration



Project & sprint planning



Product & sprint backlog management



Task development & code commit tracking



Bug / issue logging



Reporting with PDF export

Actors

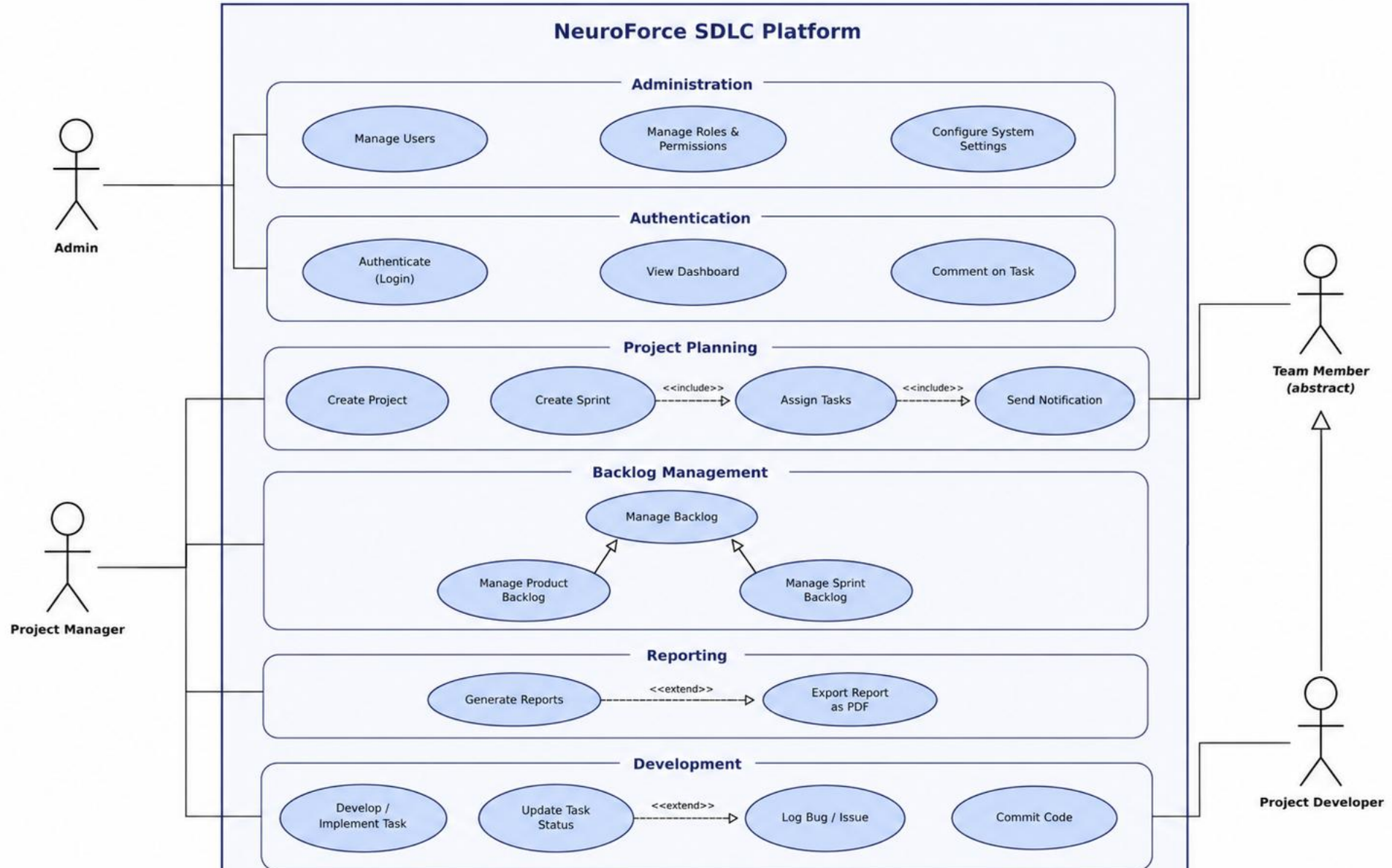
Who interacts with the NeuroForce system

Actor		Description
	Admin	Manages users, roles & permissions, and system settings.
	Team Member (Abstract)	Base actor representing every platform user; not directly instantiated.
	Project Manager	Creates projects, manages backlog, assigns tasks, generates reports.
	Project Developer	Implements tasks, updates status, commits code, logs bugs.

Why is Team Member abstract? It cannot be instantiated on its own — it defines shared behavior (login, dashboard, notifications) that Project Manager and Project Developer inherit via UML Generalization.

Complete Use Case Diagram

Actors • System Boundary • Use Cases • Association • Include • Extend • Generalization



Administration Module

Use cases owned by the Admin actor



Manage Users

Purpose:

Create, update, deactivate user accounts.

Actor:

Admin

Input:

User details (name, email, role)

Output:

Updated user record



Manage Roles & Permissions

Purpose:

Define access levels for each role.

Actor:

Admin

Input:

Role name, permission set

Output:

Role-permission mapping



Configure System Settings

Purpose:

Adjust global platform configuration.

Actor:

Admin

Input:

Setting name & value

Output:

Applied system configuration

Project Management Module

Authentication, Planning & Collaboration use cases



Authenticate (Login)

Verify user credentials to grant system access.



View Dashboard

Display personalized project & task overview.



Create Project

Set up a new project with scope & timeline.



Comment on Task

Add discussion notes on a specific task.



Create Sprint

Define a time-boxed sprint within a project.



Assign Tasks

Allocate backlog items to team members.



Send Notification

Alert assignees when tasks are allocated.

<<include>> Relationships

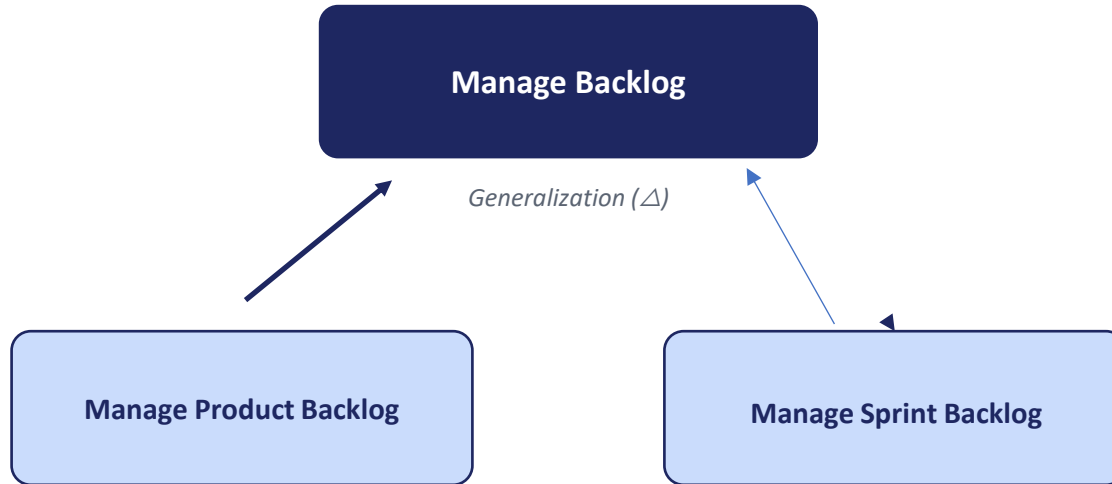
Create Sprint ► Assign Tasks

Assign Tasks ► Send Notification

Include shows mandatory reused behavior — the base use case always triggers the included one.

Backlog Management Module

Organizing and prioritizing project work



What is Generalization?

A child use case inherits and specializes the behavior of a parent use case, shown with a hollow-triangle arrow pointing to the parent.

How the Project Manager works:

- Maintains the Product Backlog — the full list of features & requirements.
- Curates the Sprint Backlog — items selected for the current sprint.
- Both specialize the general Manage Backlog capability.
- Enables prioritization of work across releases.

Development Module

Where developers build and track their work



Develop / Implement Task

Write code to fulfill an assigned task.



Update Task Status

Move a task through its workflow states.



Commit Code

Push completed code changes to the repository.



Log Bug / Issue

Record a defect discovered during development.

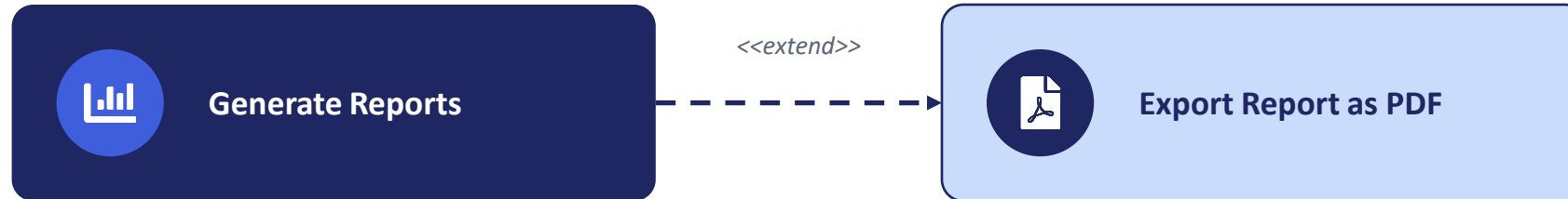
<<extend>> Relationship

Update Task Status → <<extend>> → Log Bug / Issue

Extend is used because logging a bug is an optional, conditional behavior that only occurs when a defect is found while updating status — it is not part of the base flow every time.

Reporting Module

Insights for the Project Manager



Generate Reports

Compiles project metrics such as sprint velocity, task completion, and workload — giving the Project Manager visibility into progress.

Export Report as PDF

Optionally converts the generated report into a downloadable PDF for sharing outside the platform.

Why Extend?

Exporting as PDF is optional — a manager may simply view the report on screen. Because the extension only occurs under a specific condition (the user chooses to export), <<extend>> is the correct relationship, not <<include>>.

UML Relationships

Notation used in the NeuroForce use case diagram



Association

Definition: A solid line linking an actor to a use case it participates in.

Example: Admin — Manage Users



Include

Definition: Dashed arrow with <<include>>; base use case always invokes the included one.

Example: Create Sprint <<include>> Assign Tasks



Extend

Definition: Dashed arrow with <<extend>>; optional behavior added under a condition.

Example: Update Task Status <<extend>> Log Bug/Issue



Generalization

Definition: Solid line with hollow triangle; child inherits parent's behavior.

Example: Manage Product Backlog △ Manage Backlog